Problem 16.15

A $(2n-1)$-payment annuity-immediate has payments

$$1, 2, \ldots, n-1, n, n-1, \ldots, 2, 1.$$

Show that the present value of the annuity one payment period before the first payment is

$$a_{\overline{n}|}\ddot{a}_{\overline{n}|}.$$

Solution

Let

$$v = \frac{1}{1+i}.$$

Since the annuity is an annuity-immediate, the first payment is made one period after the valuation date. Therefore, the present value is

$$\begin{aligned} PV &= v + 2v^2 + 3v^3 + \cdots + (n-1)v^{n-1} + nv^n \\ &\quad + (n-1)v^{n+1} + (n-2)v^{n+2} + \cdots + 2v^{2n-2} + v^{2n-1}. \end{aligned}$$

Now consider the product

$$a_{\overline{n}|}\ddot{a}_{\overline{n}|}.$$

Using

$$a_{\overline{n}|} = v + v^2 + \cdots + v^n$$

and

$$\ddot{a}_{\overline{n}|} = 1 + v + v^2 + \cdots + v^{n-1},$$

we get

$$a_{\overline{n}|}\ddot{a}_{\overline{n}|} = (v + v^2 + \cdots + v^n)(1 + v + v^2 + \cdots + v^{n-1}).$$

Expanding this product, the coefficient of v^k increases from 1 to n and then decreases from $n-1$ to 1:

$$\begin{aligned} a_{\overline{n}|}\ddot{a}_{\overline{n}|} &= v + 2v^2 + 3v^3 + \cdots + (n-1)v^{n-1} + nv^n \\ &\quad + (n-1)v^{n+1} + (n-2)v^{n+2} + \cdots + 2v^{2n-2} + v^{2n-1}. \end{aligned}$$

This is exactly the present value of the given annuity. Hence,

$$PV = a_{\overline{n}|}\ddot{a}_{\overline{n}|}.$$

Therefore, the present value one payment period before the first payment is

$$\boxed{a_{\overline{n}|}\ddot{a}_{\overline{n}|}}.$$